提供所用材料/样品/设备的来源

is commercially available	was collected	was performed (by/in)
was a kind gift (from)	was devised	was provided (by)
was acquired	was/were found (in)	was purchased (from)
was carried out	was generated	was supplied (by)
was chosen	was modified	was used as received
was conducted	was obtained	was used as supplied

下面是这些词汇或短语应用的例子：

- **A three-step approach was used** to determine the muscle force.
- The extracts **were obtained by using** two different methods.
- The fetal bovine serum used in this study **was provided by** Gibco, UK.
- Dextran standard 5000 **was purchased from** Sigma Aldrich.
- A laboratory-scale fixed-bed reactor **was used in this study**.
- The cell lines **were a kind gift from** Dr David Louis (Massachusetts General Hospital, MA, USA).
- The inclusion criteria used here are **a modified version of** those in a previous study[22].
- **All** volumetric lung images **were acquired from** healthy asthmatic subjects.
- **In the majority of the tests**, the chickens were kept in a temperature test chamber.

提供必要的背景信息和/或描述所使用的材料/样品/设备

你可能需要详细描述你使用或构造的设备/工具或器械，或者描述照片或图表，以便读者可以直观地看到它。为了准确地做到这一点，你需要使用多种语言来描述空间关系、位置和地点，例如：

Above	over	on top (of)	**Verbs**
adjacent	adjoining		align
across	along	against	arrange
at the front	on the front	in the front	assemble
below	under	underneath	attach to
beside	alongside		bisect
boundary	edge	border	connect
circular	rectangular	conica	converge
downstream	upstream (of)		couple embed
equidistant	equally spaced		encase
exterior	interior		enclose
in the vicinity	in close proximity	nearby	fasten fit
inside	within		fix install
lateral	sideways	horizontal	intersect
on each side	on either side	on both sides	join
on the right/left	to the right/left		locate

opposite	facing	mount
out of range	within range	orient place
parallel (to/with)	perpendicular (to)	position
symmetrical	asymmetrical	situate space surround

下面是这些词汇或短语应用的例子：

- In the experiment, source and detector **were positioned** on opposite sides.
- The apparatus consists of **a circular tube fitted with** rectangular-winglet tape (RWT) vortex generators.
- The compression axis **is aligned with** the rolling direction…
- The source light is polarised **horizontally** and the sample beam can be scanned **laterally**.
- The microtubules that **were embedded in** collagen gel were **oriented perpendicular to** the direction of the magnetic field.
- Measuring stations **were spaced** 20 cm apart, starting 2 cm **downstream of** the pipe entrance.
- A scanner **was mounted on top of a** metal structure, while pots were placed **underneath, over** a cart moving at constant speed **in the direction perpendicular to** the line of acquisition.

在下面哪个句子中，桌子与墙的距离最近？

The table was placed	against the wall.
The table was placed	alongside the wall.
The table was placed	beside the wall.
The table was placed	flush with the wall.
The table was placed	in contact with the wall.
The table was placed	next to the wall.
The table was placed	right against the wall.

在下面哪个句子中，时钟与门的距离最近？

The clock was positioned	just above the door.
The clock was positioned	slightly above the door.
The clock was positioned	immediately above the door.
The clock was positioned	directly above the door.
The clock was positioned	right above the door.

提供关于材料和方法的精确的语言

（ⅰ）动词

方法部分中的动词分为三类。第一类包括与学术研究相关的常用动词，例如：attempt、consider、conduct、determine、investigate、report和verify。这些能在“附录B：研究中的动词”部分找到。第二类包括特定领域的专门的动词，例如：anneal、clone、dissect、ionise和infuse。这些动词未包含在下面的表格

中，因为它们在其他领域是没有用的。下面的表格包含第三类动词的示例：在不同领域中用于说明具体做了什么或用了什么的技术性的动词。

was adapted	was divided	was operated
was added	was eliminated	was optimised
was administered	was employed	was performed
was adopted	was estimated	was placed
was adjusted	was evaluated	was plotted
was altered	was examined	was positioned
was analysed	was excluded	was prepared
was applied	was exposed	was processed
was arranged	was extracted	was produced
was assembled	was fabricated	was quantified
was assessed	was filtered	was recorded
was assumed	was formulated	was recovered
was attached	was generated	was regulated
was calculated	was immersed	was removed
was calibrated	was implemented	was repeated
was carried out	was included	was represented
was characterised	was incorporated	was restricted
was collected	was initiated	was retained
was combined	was input	was retrieved
was compared	was inserted	was sampled

was computed	was installed	was scored
was conducted	was inverted	was selected
was connected	was isolated	was separated
was constructed	was located	was simulated
was controlled	was maintained	was solved
was converted	was maximised	was stabilised
was created	was measured	was substituted
was defined	was minimised	was synthesised
was derived	was modelled	was tracked
was designed	was modified	was transferred
was determined	was monitored	was treated
was discarded	was normalised	was varied
was distributed	was obtained	was utilised

（ii）顺序语言

对流程中每个步骤的时间和顺序进行精确的描述，使读者和同行审稿人能够对其进行想象是至关重要的，如果他们愿意的话可以重复该流程，并将其结果与你的结果进行比较。then（然后）或next（下一步）之类的词描述了事件的顺序，但是它们并没有提供有关每个步骤花费了多长时间或下一步要多久的信息。要提供这些信息，你需要一个更精确的顺序词汇表。在描述结果时，顺序语言同样重要。

在STEMM写作中，表达顺序的词汇和短语可以分为八组，如下所示：

第一组 包含提及在开始实验和模拟之前，或开始观察结果之前发生事件的词汇和短语：

It was apparent beforehand that a reduction in temperature would be a desirable outcome.

第二组 标记实验/模拟的开始，或你正在描述的第一个步骤/结果：

At the beginning the temperature was stable, as predicted.

第三组 包含词汇和短语，可以告诉你事件发生的顺序，但不提供有关时间顺序的信息：

The temperature was increased to 49℃ and then reduced to 30℃.

温度的下降可能在温度达到49℃后不久发生，或者可能在很长时间后发生。then这个词只会告诉读者事件的顺序。

第四组 用于表达两个事件之间（仅仅）是很短的时间，或者该事件在短暂的等待期之后发生：

The temperature increased to 49℃ but soon dropped to 30℃.

第五组 表达事件之间的时间间隔很长，或者正在描述的事件发生在序列的末尾：

The temperature was increased to 49℃ and later reduced to 30℃.

第六组 是非常有用和重要的。它包含表达两个或多个事件在同一时间/同一时期发生的词汇和短语，或者一个事件与另一个事件恰好在同一时刻开始，即这两个事件相互关联。

有时使用此分组中的项目来传达事件之间可能的因果关系或相关性：

The temperature dropped sharply when we reduced the pressure.

第七组　标记顺序的结尾:

At the end there was a noticeable drop in temperature.

第八组　提及在你完成实验/模拟之后或在你观察结果之后发生的事件:

At the end there was a noticeable drop in temperature but it was decided afterwards to omit this from the input data.

将下面的词汇/短语正确分组:

after	earlier	in time	secondly
afterwards	eventually	initially	shortly after
as	finally	instantly	simultaneously
as soon as	firstly	lastly	soon
at once	followed by	later	straight away
at first	following	later on	subsequently
at that point	formerly	meanwhile	then
at the beginning	immediately	next	to begin with
at the end	in advance	once	to start with
at the same time	in due course	originally	towards the end
at the start	in the beginning	previously	until
beforehand	in the end	prior to	when
directly	in the meantime	quickly	while

答案

第一组：在开始之前

beforehand earlier	formerly in advance	originally previously	prior to initially

第二组：在开始的时候/第一步

at first at the beginning	at the start firstly	in the beginning initially	to begin with to start with

第三组：步骤/顺序

after afterwards earlier	followed by following formerly	next previously prior to	secondly (etc.) subsequently then

第四组：过了一会儿

quickly	shortly after	soon

第五组：在后期或接近末期，等了一会或等待一段时间后

eventually in due course	in time later later on	subsequently towards the end

第六组：几乎在一个期间或者完全同时发生

as	immediately	simultaneously
as soon as	instantly	straight away
at once	in the meantime	until
at that point	meanwhile	when
at the same time	once	while
directly		

第七组：结尾/最后一步

at the end	finally
eventually	in the end
	lastly

第八组：结束之后

afterwards	later
eventually	later on
	subsequently

证明选择的正确性

这里包括引入你做出某种方法或程序上的选择的原因的短语，例如 in order to 和一张“快乐的”词汇表（‘happy’ words）以及表达所做选择的优点的动词。

by doing..., we were able to	**'happy' words ☺**	**Verbs**
chosen for/to		
designed for/to	accurate	achieve
for brevity	consistent	allow
for convenience	direct	avoid
for maximum effect	easy	compensate (for)
for (the sake of) simplicity	excellent	confirm
for the following reasons:	important	determine
in an attempt to	precise	enable
in order to	relevant	enhance
offer a means of	robust	ensure
our aim was to	satisfactory	establish
provide a way of/to	simple	facilitate
selected on the basis of	suitable	guarantee
so as to	useful	improve
so/such that		include
the advantage of		increase
the reason for		limit
thereby	**CAUSE and RESULT signalling connectors** (参见第1.5.2节“表示原因”“表示结果”部分)	minimise
thus		obtain
to take advantage of		overcome
to this end		permit
which/this allowed		prevent
which/this permitted		provide
which meant that		reduce
with the aim of		remove
		simplify
		validate

这是一些如何使用它们的例子：

- **For brevity**, these equations are not included in this paper.
- **To** make the problem tractable, we reduced the number of possible scenarios.
- Zinc oxide was drawn into the laminate **with the intention of enhancing** delaminations and cracks.
- A highly conductive filler is added to the matrix **in order to ensure** electrical conductivity above the required level.
- **The advantage of** using three-dimensional analysis was that the out-of plane stress field could be obtained.
- By partitioning the array, **we were able to** identify all the multipaths.
- **For simplicity**, only a single value was analysed.
- The samples were dried before being weighed, **so allowing** accurate calculation of the ratio of soil to stones.
- **For verification purposes**, tests were repeated for 9 different combinations of temperature and strain rate.
- The survey was administered during the spring, **which avoided problems of** low activity during the summer months.
- Our control drug was given to all patients, **therefore we were able to** separate the effects of increased pH from other factors.

表明你对待研究的谨慎态度

下面表格中的词汇大多是副词形式（例如accurately），但它们也可以形容词形式出现（例如：accurate）。

accurately	entirely	immediately	rigorously
always	every/each	independently	separately
appropriately	exactly	individually	smoothly
at least	firmly	never	strictly
both/all	frequently	only	successfully
carefully	freshly	precisely	suitably
completely	fully	randomly	tightly
constantly	gently	rapidly	thoroughly
correctly	good	reliably	uniformly
directly	identical	repeatedly	vigorously

这是一些如何使用它们的例子:

- Care was taken to maintain **strict** anaerobic conditions during extract preparation.
- These cells were then washed **thoroughly at least** three times.
- To prevent an emulsion from forming, only **gentle, repeated** inversion was used.
- The specimen was monitored **constantly** for 24 hours.
- **Frequent** readings were taken to update the stress conditions **smoothly**.
- The specimen was **tightly** clamped at two corners to minimize any unexpected gaps.
- After irradiation, the reaction mixture **was carefully** condensed.
- 6 **pristine** specimens **were handpicked** under a microscope.

- We used a **reliable** analytical method based on high-performance liquid chromatography.

比较或者参考其他的研究的材料/方法

选项1：你使用的材料或方法与你引用的研究**完全相同**。

according to	as reported by/in	given by/in
as described by/in*	as reported previously	identical to
as detailed by/in	as suggested by/in	in accordance with
as explained by/in	can be found in	previously shown in/by
as in	described elsewhere	the same as that of*/in
as proposed by/in	details are given in	using the method of/in
	following x *et al.*	

*注：在引用时，**by** 和 **of** 后面通常是研究者或者团队的名字（例如 as described by Ross 或 using the method of Ross *et al.*），而**in** 后面通常是作品（例如 as described in Ross *et al.* [2020]）。

另外一种选择是仅仅在句子的合适位置提供参考文献："We used the Shapiro-Wilk test[11] to determine whether a given sample was from a normal population."

选项2：你使用的材料或方法与你引用的研究是**类似**的。

a (modified) version of	largely the same	(to) adapt
adapted from	more or less identical	(to) adjust
almost the same as	partly based on	(to) alter
based on	practically the same	(to) change

essentially identical	similar (to)	(to) modify
essentially the same	slightly modified	(to) refine
except for/that	virtually the same	(to) resemble
following x *et al.*	with some adjustments	(to) revise
in essence	with some alterations	(to) vary
in line with	with some changes	
in principle	with some modifications	

选项3：你使用的材料或方法与你引用的研究**有本质差别**。

a novel step was…	CONTRAST	(to) adapt
adapted from	signalling	(to) adjust
although in many ways similar	connectors, e.g.:	(to) alter
although in some ways similar		(to) change
although similar to		(to) modify
based on	however	(to) refine
except for/that	whereas	(to) resemble
instead (of)	by contrast	(to) revise
loosely based on		(to) vary
partly based on		
unlike		
with the following modifications:		

注：如你看到的，大多数词汇也可以用在选项2或者选项3中。在选项2中使用它们的时候，你可能不需要说明你所使用的程序/材料与你所引用的文献中所使用的程序/材料之间的区别。在选项3中，你应该确切地指出这些明显的差异，**特别是当你的研究改进了原有文献中的方法时**。

下面是这些词汇或短语应用的例子：

- The method was **essentially the same as** that previously described elsewhere[5] for use in *X. tropicalis*.
- **We follow** Sobral *et al.* (2013) and apply one magnitude of dust extinction to all of our HAEs.
- The analytical method **was adapted from** British Pharmacopeia[7].
- The method used was **identical to** that previously reported in ref. 17.
- Activated calf thymus DNA was prepared **as decribed by** Pedrali-Noy and Weissbach (17).
- We **modified** the Du and Parker filter to address these shortcomings and we refer to this modified filter as the MaxCurve filter.
- Data were collected using **a revised version** of the Portrait Values Questionnaire (9).
- However, **in this study, instead of using** a coil around the membrane cell, we placed the coil on top of the cell.

表明问题出现在什么地方

minimise problem (no big deal)	**minimise responsibility (not my fault)**	**maximise good aspects**
did not align precisely	as far as possible	acceptable

immaterial	impossible	fairly well
it is recognised that	impractical	quite good
less than ideal	inevitably	reasonably robust
minimal	(it was) difficult to	
minor deficit	(it was) hard to	
negligible	limited by	
not identical	necessarily	**refer to a possible**
not perfect	not possible	**solution:**
not significant	problematic	Future work should…*
only approximate	unavoidable	Future work will…*
rather time-consuming	unworkable	
slightly disappointing		
slightly problematic		
unimportant		
(very) small		

*注：短语 future work should 和 future work will 之间有一个重要的差别。当使用 future work should 时，表示你在建议一个未来的研究方向，并且邀请同领域的研究人员接受这个挑战并做出研究成果。而当你用 future work will 时，你是在把自己的计划和意愿传达给同领域的人，而未来的工作将传达作者自己的计划、意图或当前正在进行的工作。

下面是这些词汇或短语应用的例子：

- Despite our routine of monthly examinations, follow-up **was not perfect**.
- It **was slightly difficult to** deposit a uniform film of the

liquid, probably owing to its low viscosity.

- Although some duplication **was unavoidable**, this was **minimized by** the frequent use of cross references.
- The probability is **very small**, and hence we believe such contaminants have **only a negligible effect on** this study.
- Varying the diameter required the machine to be recalibrated, which **was rather time-consuming**.
- Although centrifugation did not remove all the excess solid drug, the amount remaining was **negligible**.
- Due to the lack of multiwavelength data it was **not possible** to apply additional broad-band selections.
- **Inevitably**, considerable computation was involved.
- This procedure **was slightly problematic** in that the program reported successful setup even if a nonexistent file was specified.

2.5　语言和写作技巧

这一节涉及方法部分重要的三个语言领域：

- 所有权：动词时态和无施动者的被动语态
- 介词
- a 和 the 的用法

2.5.1 动词时态和无施动者的被动语态

在写方法部分之前，查看你的目标文章以及目标期刊的作者指南，以弄清楚该文章主要是被动语态还是主动语态。尽管主动语态（we investigated/we conclude）在研究写作中很常见，方法部分的一些过程通常仍会以被动式表达。

科技写作者通常使用无施动者的被动语态（was/is found），而不是被动语态加施动者（was/is found **by us**）的形式，因此在大多数情况下，实际执行动词的人不会被提及。这会造成引起歧义的高风险，因为无论是描述你自己的工作（samples were collected using a sterile swab）还是其他研究人员的工作（samples were collected using a sterile swab），无施动者的被动对象看起来都是完全相同的。这种歧义性会严重影响你的研究论文的成功，因为它可能使你失去对论文工作的所有权，并不知不觉地将全部或部分工作归功于另一项研究。

第二个潜在的歧义来自方法部分中的动词时态的使用。过去时通常用于报告你所做的事情，而现在时通常用于描述标准程序或设备。因此，被动语态中的动词时态错误可能会改变信息的含义，例如，你自己的工作可能会与你描述的标准程序混淆：

报告你做了什么（一般过去时无施动者的被动语态）

A flexible section **was** inserted in the pipe.

描述标准程序或设备（一般现在时无施动者的被动语态）

A flexible section **is** inserted in the pipe.

必须要认真地发现和解决这样的歧义。否则的话，读者或审稿人将很难或不可能知道你指的是自己的工作还是其他研究人员的工作或标准程序。这对你的研究论文的成功非常不利。不幸的是，你对项目的熟悉使你更难以发现可能存在歧义的地方：你所写内容的含义对你而言似乎总是明白易懂的，但这并不意味着对读者而言是一目了然的。

请记住这本书开头的关键信息之一是：**科技写作的目的不仅仅是使读者能够理解；而且是使读者不可能理解不了。**确保你自己的贡献清晰易辨的一种方法是用“本研究（In this study）”之类的语言做标记。下面是方法部分中可能需要的无施动者被动式的五种可能用法，并提供了一些使你的意思更清楚的方法建议。

	你想表达什么？	如何清晰地表达出你的意思？
1	X was (collected/ modified) **by men in the procedure or work that I carried out.**	• 转换到主动语态（We collected/ modified）。 • 添加像here/in this work/in our model 这样的词汇或短语。 • 使用“假”主语，比如：“This experiment/The procedure described above.”
2	X was (collected/ modified) **by the person whose procedure or work I am using as a basis for mine or comparing with mine.**	• 提供参考资料。 • 添加像 in their work/ in that model 这样的词汇或短语。 • 使用“假”主语，比如：“That experiment/The probe used in their study.”

3	X is (collected/modified) **normally, i.e. as part of an established or standard procedure.**	• 即使是标准的程序，你可能还是需要参考文献，这取决于它为人们所熟知的程度。 • 使用像 using standard procedures/ as in[5]这样的短语。
4	X is (collected/ modified) **as you can see in Fig. 1, and it was collected/modified by me.**	• 明确地将图像的内容标识为你自己的工作："The experimental setup used here is shown in Fig.3." • 转换到主动语态："We collect/ modify X. "
5	X was (collected/ modified) **by me, but in my field/this journal the Present Simple tense is used.**	• 转换到主动语态："We collect/ modify X. " • 添加像here/in this work/in our model 这样的词汇或短语。 • 使用"假"主语，比如："This equation/The model."

拥有自己的工作成果并将它与当前的其他工作分开是至关重要的。完成方法部分后，请检查**每一个句子**以确保动词使用正确的时态，即该动词正确表达了你想要的功能和含义，并且没有简单地假设该动词的功能和含义对你的读者是明白易懂的。

2.5.2 介词

介词对含义的表达有重要的影响。例如 evidence of something 指一个可测信号，表示证据在那里或者存在证据，而 evidence for something表示证据可能在那里，或者可能存在。有一些动

词的含义完全依赖介词：例如 X was substituted **for** Y 表示以 X 代替 Y，而 X was substituted **with** Y 表示以 Y 代替 X。

你是在为全球读者写作，一些读者可能难以理解介词的含义，因此即使英语是你的第一语言，也请务必谨慎。以下是改善介词用法的五种有用策略：

1.首先要格外注意介词

在这些经过编辑的摘录中，你能找到多少个以下的介词（of、for、with、on、in、through、at、over、from、about）？答案在每个摘录后的括号中。

- A helium-neon laser beam, having a wavelength of 633 nm, was directed onto the sample, through a chopping wheel, at an angle 59° from the normal. The specularly reflected beam was absorbed onto a black card, and the scattered light was collected at normal incidence to the sample using an X10 microscope objective. *(8)*
- The solutions were placed in the refluxing bath immediately after mixing, and the temperature of the bath was increased to 70°C in 15 min and kept there for 2 h. The solutions were then aged for 24 h at room temperature, before being diluted with an equal volume of EtOH and stirred for 10 min, to give the solution used for spin coating. *(11)*
- The film is placed in a sealed chamber on the sample stage of the ellipsometer; first dry N2 gas is passed through the

chamber to empty the pores, and after N2 has been bubbled through, the liquid adsorbate is passed over the sample to fill the pores. By assuming that all the accessible pores are completely empty or filled with adsorbate, the pore volume and index of the solid skeleton can be determined. Measurement of *nf* for both the dry and saturated films allows both *vp* and *ns* to be determined with the assumption that *np* has the same value as that of the bulk adsorbate in the saturated case, and of air (*np* = *1*) in the dry case. *(15)*

- The IR camera that we used in the setup is a Raytheon Radiance HS, and operates in the 3–5 μm spectral region. It is sensitive (with a 1 ms integration time) to surface temperature changes of ~0.03°C, and can be operated at full frame rates of 140 Hz with that sensitivity. We have also observed the effects reported here with a microbolometer focal plane array camera, operating in the long wavelength of the IR. *(11)*
- Hexagonally packed intermediate (HPI) layer from Deinococcus radiodurans, a kind gift of Dr. W. Baumeister, was extracted from whole cells (strain SARK) with lithium dodecyl sulfate, and purified on a Percoll density gradient. A stock solution (1 mg/ml protein) was stored in distilled water at 4°C. Porin OmpF trimers from E. coli strain BZ 1 10/ PMY222 (Hoenger *et al.*, 1993) solubilized in octyl-polyoxyethylene were mixed with solubilised dimyristoyl phosphatidylcholine (Sigma Chemical Co., St. Louis, MO) at a ratio (w/w) of 0.2 and a protein concentration of 1 mg/ml. 1,2-Dipalmitoyl-phosphatidylethanolamine (DPPE)

from Sigma was solubilized in chloroform:hexane (1:1) to a concentration of 1 mg/ml. The resulting solution was diluted in buffer solution (150 mM KCl, 10 mM Tris, pH 8.4) to a concentration of 100 μg/ml. *(20)*

2.注意目标研究论文中描述过程的动词+介词组合

这是摘自方法部分第2.3.2节的片段。一些介词很明显，而其他的则不太明显。你会选择这些介词吗？

Materials and methods

The MC code system PENELOPE 2003 (Salvat *et al.* 2003) and the main program PENCYL were used to calculate the absorbed dose **in** the detector and surrounding water phantom. In order to minimize the time **of** the simulations, the geometry **of** the diamond layer and electrodes was simplified **by** adopting an infinite-slab geometry [...] Since the present study focuses **on** the metal/diamond/metal interface phenomena, the housing **of** the detector was regarded **as** water equivalent and thus replaced **by** water. The influence **of** metallic electrodes of different thicknesses **on** the absorbed dose **to** the diamond was investigated. Electrodes made **of** silver were simulated first. Silver is a common material because it has a high electrical conductivity, forms ohmic contact **with** diamond and the technology **of** its thermal deposition as well as contacting the detector **with** silver epoxy glue is well established. [...]

A careful selection **of** simulation parameters was

needed in order to preserve accuracy while completing the calculations **within** reasonable CPU. **In** the course of the simulations, each particle (a primary or secondary photon, electron or positron) was transported until its energy fell **below** the corresponding pre-selected absorption energy, E_{abs}; when this happened, its energy was deposited locally and simulation **of** the particle was discontinued. The mixed algorithm implemented **in** PENELOPE **for** the tracking **of** electrons and positrons is governed **by** specific simulation parameters, namely C_1, C_2, W_{cc} and W_{cr}, whose definitions and role are explained at length **in** Salvat *et al.* (2003). For the sake of brevity, here we just enumerate them and quote the adopted values. [...]. We found it convenient to use different simulation parameters depending **on** the distance **of** the particles **to** the active layer and electrodes. To this end, we considered that the detector, where the most 'conservative' simulation parameters were selected, was surrounded **by** three 'shells' of water **with** successively increasing thickness and values **of** E_{abs}. [...].

下面是动词+介词组合的一些例子。研读你的目标文章，将你从中挖掘到的组合补充到列表里。

was applied across all cohorts	was infused within 5 minutes
was calcinated at 450°C	was input into a model
was calcined in static air	was labelled with a dye

was calibrated against real-world data	was maintained in this condition
was considered for analysis	was manufactured in mild steel
was converted into fatty acids	was manufactured with steel pins
was cultured on coated plates	was mapped against the reference genome
was cured in an autoclave	was normalised for exon length
was degassed in a vacuum oven	was performed at quasi-static rates
was degassed with argon	was performed on tissue samples
was degreased in acetone	was placed under vacuum
was divided into rectangular cells	was precipitated with isopropanol
was exposed to heat	was propagated in 1:3 ratio
was extracted from frozen samples	was propagated with Invitrogen
was extracted with a Hilbert transform	was replaced with feed gas
was fitted with a plastic tube	was retrieved from the database
was grit-blasted with grade 60 grit	was spread onto the surface
was ground with a mortar and pestle	was subjected to impact
was grown on a mineral medium	was transferred into a well plate
was incubated for 2 h	was treated with primary antibodies

3.用词义清楚的词汇代替介词

这在句子开头处特别重要，因为如前所述（第1.5.2节），句子开头的方式是其成功的关键。

与其用这个句子：

From this estimation, we ranked the search results obtained and then classified them according to size.

不如用这个句子：

Using this estimation, we ranked the search results obtained and then classified them according to size.

当你写出下面这个句子：

With the increase in computer processing speeds, the time taken for simulations has reduced.

你是指由于增加（as a result of the increase）还是在增加的同时（at the same time as the increase）?

当你写出下面这个句子：

With many attempts, we were able to position the probe in the centre of the tissue.

你是指在多次（失败）尝试之后［after many (failed) attempts］还是在某些情况下（in several cases）?

With 这个介词特别容易出“问题”。That dog **with one eye** 可以是指“这只狗用一只眼睛”（在看的时候闭上一只眼睛）或者是“只有一只眼睛”。此外，**with** 可以在某些情况下传达带有意图，而 **by** 则表示缺乏意图：用金属棒击打（being hit **with** a metal bar）是有意的，而被金属棒击打（being hit **by** a metal bar）则表明缺乏意图。在另一个示例中，如果写成“X was coated **with** Y”，则可能意味着Y是用于涂盖X的材料。如果想要表明Y用来涂覆X的过程，写成“X was coated **by** (doing) Y”则会更清楚。

4.寻找介词使用的模式

当你查看目标研究文章时，你可能会发现 using 出现得比 with 更频繁，并且by通常出现在以后缀 -ing结尾的描述过程的词或短语之前。下面是一些例子：

was analysed USING a computing cluster	was generated USING clinical covariates
was assessed USING an Instron 4507	was improved USING a shielding gas mix
was calculated USING another	was modeled USING 13 elastic elements
was carried out USING a Ministat 251	was mounted USING adhesive tape
was computed USING a flow algorithm	was optimised USING Bayesian criteria
was conducted USING an optical microscope	was performed USING an X-ray diffractometer
was controlled USING an automatic system	was performed USING a mapping algorithm
was designed USING an eArray platform	was quantified USING the phase-locking factor
was developed USING a laser	was removed USING a grinder
was evaluated USING an analog scale	was reported USING the HGVS nomenclature
was evaporated USING a rotary evaporator	was selected USING a filter
was extracted USING a reagent solution	was solved USING the penalty method

was filtered USING a Butterworth filter	was sonicated USING a Branson Sonificator
was achieved BY electroplating	was extracted BY applying a filter
was assessed BY analysing…	was formed BY incorporating…
was assessed BY angiography	was initiated BY administering a drug
was calculated BY dividing…	was precipitated BY adding…
was confirmed BY immunostaining	was prepared BY chemical vapour deposition
was determined BY measuring ATP levels	was prepared BY dissolving…
was differentiated BY withdrawal of bFGF	was replaced BY bFGF

5.避免使用一连串的介词短语

连接在一起的介词短语往往会相互“感染”，导致句子混乱。例如在这个句子中：“He gave a lecture about liver cancer at the hospital last January.”不确定“在医院”限定的是讲座还是肝癌，也不确定是讲座还是癌症病例发生在一月份。考虑将句子重新组合为：

Last January he gave a lecture at the hospital; the subject was liver cancer.

像下面这种带有多个介词短语的句子，对于读者来说更加难以理解：

The tray **with** the samples was placed **in** the oven **at** 250°C

with protective gloves to avoid injury **for** one hour.

手套也在烤箱里吗？难道目的是避免一个小时的伤害？考虑将句子重新组合为：

The tray **containing** the samples was placed in the oven for one hour **at** 250℃. Protective gloves were worn to avoid injury.

同样，很难在如下的句子中发现其新颖之处：

This is the first study to use X-ray imaging **over** a period of five years to measure contact angles **within** oil-bearing rocks **at** reservoir conditions.

这是第一个使用X光成像的研究吗？是第一个在五年内使用X射线成像的吗？第一个使用它来测量含油岩石内部的接触角的方法吗（measure contact angles within oil-bearing rocks）？第一个在水库条件下（reservoir conditions）做到这一切的人吗？你总是很清楚句子的意思，但这并不意味着每个读者都明白。

2.5.3　使用A/AN（不定冠词）、∅（零冠词）和THE（定冠词）

冠词的使用不是仅由语法决定的，冠词具有深刻的表达功能，作者经常要在合乎语法规范的冠词之间进行选择，以便做

出清晰的表达。在一个句子中有可能使用**the**或**a** 都是正确的，但是正确并不意味着句意完全相同。因此，尽管冠词的使用有其语法规则，但上下文在决定使用**a/an**、∅还是**the**时通常起着重要作用。这在高科技和复杂信息的表达中尤为重要，在这种情况下，冠词的选择有时取决于作者想要表达的内容，有时取决于作者所假设的与读者共享的知识的水平。

一个很好的开始的规则是：单数可数名词必须带有冠词（例如**a**或**the**）。但应用这一规则就像应用语法规则一样，并不是直截了当的。首先，你还不清楚哪些名词是可数的，即使你知道一个名词是可数的，你又怎么决定是用 **a** 还是 **the** 呢？

本节将考虑以下三个问题：

可数名词到底是什么？
我应该用**a/an**、∅还是**the**呢？
当**a/an**和**the**都能用时，它们在含义上有区别吗？

问题1：可数名词到底是什么？

可数名词是可以形成复数的名词。有些名词是可数的，有些名词是不可数的，有些名词既是可数的又是不可数的，取决于它们的用法。举例来说，许多所谓的不可数名词，有时候也可以以复数的形式出现，比如 death 或 childhood：

There have been three **deaths** this year from pneumonia.

Our **childhoods** were very different; I grew up in France and she grew up in China.

同样地，像 industry 这样的名词也有可能以复数的形式出现：

Many **industries** rely on fossil fuels.

因此当使用像 industry 这样的单词的时候，停下来思考一下，你指的是总体上的工业（不可数）还是一个特定的行业（可数）？此外，通常被认为是不可数的材料，例如混凝土（concrete）和钢（steel），在被该领域的专家使用时可以当可数名词：

We have developed **steels** containing only elements that produce certain radioactive isotopes.

这是在STEMM内容中有用的不可数名词的列表。尝试找出也可以形成复数形式的单词，然后在后面**答案**部分的列表中核对你的答案。

absence	earth	industry	protection
advice	economy	information	purity
age	education	insurance	quantity
agriculture	electricity	intelligence	reality

aid	energy	knowledge	research
air	environment	life	risk
analysis	equipment	light	safety
atmosphere	evidence	loss	salt
behaviour	existence	machinery	sand
blood	experience	noise	science
business	failure	nutrition	strength
calculation	fear	oil	stuff
cancer	fire	organisation	technology
capacity	food	oxygen	temperature
childhood	fuel	paper	transport
concrete	harm	philosophy	treatment
danger	health	physics	trouble
death	heat	policy	truth
democracy	height	pollution	velocity
design	help	power	vision
disease	history	pressure	waste
distribution	independence	progress	water

答　案

在下面的列表中，以不可数形式使用的名词（即不具有复数形式）用**粗体**表示。如果要以复数形式使用这些名词，需要添加另一个名词，例如：**items** of equipment、**methods** of transport、**types** of evidence。

absence	**earth**	industry	**protection**
advice	economy	**information**	purity
age	education	**insurance**	quantity
agriculture	**electricity**	**intelligence**	reality
aid	energy	**knowledge**	**research**
air	environment	life	risk
analysis	**equipment**	light	**safety**
atmosphere	**evidence**	loss	salt
behaviour	existence	**machinery**	sand
blood	experience	noise	science
business	failure	**nutrition**	strength
calculation	fear	oil	**stuff**
cancer	fire	organisation	technology
capacity	food	**oxygen**	temperature
childhood	fuel	paper	**transport**
concrete	**harm**	philosophy	treatment
danger	**health**	**physics**	trouble
death	**heat**	policy	truth
democracy	height	**pollution**	velocity
design	**help**	power	vision
disease	history	pressure	waste
distribution	**independence**	**progress**	water

问题2：我应该用a/an、∅还是the呢？

你可能已经知道**a/an**是泛指的，**the** 是特指的，但是在这个句子中：

There is **a** book on the shelf above my desk.

很明显，这里的 a book 指的是一本具体的书，事实上，该句子的目的就是指定或标识该书。看来，泛指和特指这一标准不能帮助你选定冠词，那你该怎么做呢？

a/an的使用

（ⅰ）首次提及单数可数名词时使用a/an:

I bought **a** cheese sandwich and **an** apple. **The** sandwich was OK but **the** apple had **a** little worm in it.

尽管两个句子都指相同（特定）的奶酪三明治和苹果，但第一个使用**a**，第二个使用**the**。这是因为作者第一次提到三明治奶酪或苹果的时候，只有他/她知道是它们。但是在第二次，作者和读者都分享了这一知识。因此，切换到**the**不是基于语法，而是基于上下文，并且使用“the”反映了读者和作家共享知识的事实。共享知识的概念与研究写作高度相关。

（ⅱ）你不知道或读者不知道你指的是哪个人/事时使用a/an:

Bring me **a** pen please.（用哪支笔并不重要）

She works in **a** bank.（我/你不知道是哪个银行）

The subject spoke to **an** interviewer.（我知道哪个但你不知道）

（ⅲ）使用a/an对单数形式的可数名词做宽泛的说明：

A semiconductor can conduct electricity under certain conditions.

∅零冠词的使用

（ⅰ）使用∅对复数形式的可数名词做宽泛的陈述：

∅ Semiconductors can conduct electricity under certain conditions.

（ⅱ）使用∅对不可数名词做宽泛的说明：

∅ Pollution is generally the result of human activity.

（ⅲ）首次提及可数名词时，请使用∅：

We conducted **∅** DNA arrays on **∅** silicon chips.

（ⅳ）使用∅表示完全不可数的名词或兼有不可数的名词：

These acids may undergo **∅** oxidation.

We found many applications in **∅** industry.

the 的使用

（ⅰ）当确实只有一种可能性时使用the：

The opening was located in **the** centre of each mesh.

We report the discovery of **the** smallest possible carbon nanotube.

（ⅱ）在读者明确知道的时候使用the：

They arranged to meet in **the** cafe.（他们通常会面的咖啡馆）

Did she get **the** job? （我们俩都知道她想要的工作）

The cheese sandwich was very good.（我之前提到的芝士三明治）

（ iii ）对读者显而易见的时候使用the，即使是没有提及过的事物或人：

I bought a new computer but **the** keyboard was faulty.

这就是共享知识的概念变得重要的地方。一提到计算机，作者和读者就可以安全地假定“共享”了键盘这一知识。准确而敏锐地确定出你可以安全地假定为共享信息的内容，对于文本的成功至关重要。

问题3：当a/an和the都能用时，它们在含义上有区别吗？

在某些情况下，含义确实有区别。例如：

1. The nodes should be attached to ∅ two receptor sites.

2. The nodes should be attached to **the** two receptor sites.

句1中有**很多**受体位点，然而句2中**只有两个**受体位点或者是**前面确定的**受体位点。这类似于X是Y的一个原因（许多原因之一）和X是Y的原因（唯一原因）之间的区别。下面是一个类似的例子：

1. This effect may hide **a** connection between A and B.

2. This effect may hide **the** connection between A and B.

在句1中A和B之间**可能有**联系，而在句2中，A和B之间**存在**已知的联系。

在其他情况下，含义没有差异，但是其选择反映了读者的假定知识。例如：

1. ∅ Improvements in virus-detecting software have recently become very important.

2. **The** improvements in virus-detecting software have recently become very important.

句1可能出现在全国性报纸上，它的读者可能不知道这些“改善”。句2可能出现在一个计算机期刊上，假定读者与作者有高水平的共享知识，因此他们会熟悉这些“改善”。

为了说明共享知识的作用，请考虑在下面的句子中使用**a**还是**the**：

a) You were booking a plane ticket on the telephone.

b) You were booking a plane ticket while sitting in the travel agent's office.

I'm going to have ____ baby next month, so I would like a very comfortable seat.

共享知识的概念对于期刊出版以及对研究论文内容的跨学科获取都具有重要意义。如果你将文章提交给拥有广泛读者的期刊，你不能假设你的关键术语和概念将被所有读者熟知，因此你在首次提及它们时可能需要使用 **a/an**，并且需要定义或解释它们。另外，如果你的领域非常狭窄，并且确定所有读者都会熟悉你的关键术语和概念，那么使用**the**可能是可以接受的。

现在查看目标研究文章中的一些句子，并使用上述所有信息，找出作者为什么在每一种情况下选择使用 **the**、**a**或 ∅。

2.6 编写方法部分

本单元介绍了方法部分的通用模板以及通常用于生成可读的、有效的文本的语言。在下面的任务中，你将通过综合本单元中的所有内容（模板，语言和写作技巧以及词汇表）来编写方法部分。

2.6.1 编写方法部分

下面任务的目的是让你学习如何描述你所做的工作和使用过的内容，以便任何读者都可以准确地重复你所做的工作，理解原因并获得相似的结果。这就要求你清楚地认识到自己对方法的了解与潜在读者之间的区别。你对方法的熟悉程度使你很难看到潜在的与语法相关的歧义性，例如使用无施动者的被动式。

要完成此项任务，想象你正在写一个研究项目，该项目是有史以来第一次尝试烹饪鸡。你可以想象直到现在人们都生吃鸡，你的任务是写一个烹饪鸡肉的食谱，把这个食谱当作研究论文的方法部分。例如，与其从“把鸡切成四份”这样的句子开始，你也许可以从整个过程的综述开始，或者从说明鸡肉的来源开始。鸡是在超市买的吗？是实验室设备提供的吗？

你需要说明用什么把鸡切开的，用斧头和用四厘米的萨巴提钢刀切的效果会很不一样！你不该这样写："Now put the chicken pieces in a hot oven for about an hour and a half.（现在把鸡放入热烤箱中大概一个半小时。）"你应该这样写："Each sample was then placed on a 300 × 600 mm stainless steel sheet and heated in a Panasonic E458X500 w standard fan-assisted oven for 94 minutes at 350℃.（每个样本放在一张300 × 600毫米的不锈钢盘上，并且在松下牌的E458X500瓦标准的风扇辅助的烤箱里以350摄氏度加热90分钟。）"

你的论文题目是"An approach to the preparation of chicken using heat and flavouring"。论文的引言部分如下：

Introduction

Chicken preparation techniques are used in a range of applications both in homes and in restaurants. Chicken is easily available and can be locally produced in most areas; in addition, it is easily digested and low in calories[1].

Since Dundee's pioneering work reporting the natural method of chicken preparation (Dundee *et al.*, 2008) in which the chicken was killed and then eaten raw with salt, there have been significant innovations. Much work has been carried out in France in relation to improving the method of slaughtering chickens[2], whereas in the USA researchers have concentrated on improving the size of the bird[3,4]. The natural method is widely used since the time required for the process

is extremely short; however, some problems remain unsolved. The flavour of chicken prepared using the Dundee method is often considered unpleasant[5] and there is a well-documented risk of bacterial infection resulting from the consumption of raw meat[6,9,11].

The aim of this study was to develop a preparation method that would address these two problems. In this report, we describe the new method, which uses seasoning to improve the flavour while heating the chicken in order to kill bacteria prior to eating.

现在来写这篇论文的方法部分。如果你一时不知如何往下写，运用模板和语言列表来帮助你前进。直到写完之后再看答案。

2.6.2 答案

下面是上面练习的答案，留意在每个句子中呈现了模板的哪一部分。

An approach to the preparation of chicken using heat and flavouring

Method

Two chicken samples were tested to determine the impact

of seasoning and heating on flavour and bacterial levels. Flavour was assessed by a testing group selected from those in a previous study[6]. Bacterial counts were determined before and after the heating process using an immunoassay system (Saliva-Check MUTANS) based on monoclonal antibody technology[11].

Two 4.5 kg frozen organic chickens were purchased from Buyrite Supermarket. Buyrite only sells grade A chickens approved by the Organic Farmers Association, thus ensuring the homogeneity of the samples. Two seasoning mixtures, Tri-8-00 and Tri-8-001, were obtained from SeasonInc® and used as supplied.

The chickens were defrosted in a sealed container at room temperature for 5 h. They were then cleaned according to the Hanks method[7], in which chickens are immersed in boiled water cooled to a temperature of 20°C, and subsequently rinsed thoroughly in an isotonic salt solution. In order to obtain two samples of equal size and weight for testing, the chickens were first skinned using a standard BS1709 Skin-o-matic. This made selecting suitable sample locations visually simpler, and enabled precise measurement of the samples prior to cutting. A 3 cm cube weighing 100 g was cut from the identical location on each chicken, using a 4 cm steel Sabatier knife[8] at a pressure of 2 psi.

The two cubes thus obtained were placed in a Phillips R2D2 Dehydrator for 10 minutes to evaporate the remaining salt solution. Immediately after removing the cubes from

the dehydrator each cube was completely spray-coated with 15 g of one of the seasoning mixtures, using an application technique similar to that described in Duck *et al.*[9] The samples were left to stand on a glass plate for 30 minutes at room temperature (16°C) to enhance absorption of the seasoning prior to heating. It should be noted that the seasoning quantities were measured using standard domestic kitchen scales, and are therefore only approximate.

Each cube was then placed on an ovenproof dish and transferred to a pre-heated Panasonic Model 33KY standard electric fan-assisted oven at 150°C for 10 minutes. The samples were then removed from the oven and allowed to come to equilibrium, after which they were assessed using the Bacteriometer test developed by Vira[9]. Flavour was assessed by the testing group using the scale described in Buds *et al.*[6].

练习5 概述练习。在下面的摘录中，为以下内容画线或做标记：模件组件、有用的介词、有趣的动词时态和顺序语言。答案见后面。

Nanoparticle scaffolds for syngas-fed solid oxide fuel cells[n]

Experimental

Fig.1 summarises schematically the formulation and fabrication procedures used in the paper. Inks were formulated from a

mixture of commercial gadolinia-doped ceria ($Ce_{0.9}Gd_{0.1}O_{1.97}$, CGO, particle size=0.5 μm) from Praxair, nanoparticle CGO ($Ce_{0.9}Gd_{0.1}O_{1.97}$, particle size=5 nm from TEM) produced by continuous hydrothermal flow synthesis (CHFS), polymer microbeads (6 μm, Microbeads AS, Spheromers CA6), terpineol as a solvent, Hypermer KD15 as a dispersant and an ethylcellulose binder (Hercules EC N-7). The CGO nanopowders were produced by continuous hydrothermal flow synthesis using a pilot plant at University College London as described elsewhere.[17] CHFS is an emerging and scalable method of preparing nanoparticles and has previously been used to make samaria-doped ceria,[3] ceria-zirconia solid solutions,[18,19] an entire phase diagram of 66 ceria–zirconia–yttria samples[20] and a range of other metal oxides[21,22] and phosphates[23] with typically sub 50 nm diameter. In CHFS, an aqueous mixture of cerium ammonium nitrate and gadolinium nitrate solution was continuously mixed with a stream of superheated water at 450 °C and 22.1 MPa in a patented confined jet mixer.[24] Particles were then cooled (*via* a pipe cooler) and then collected as an aqueous slurry at the exit of the back-pressure regulator. The particles were recovered by centrifugation and washing (three times) in an equal volume of water until the final solid sludge was obtained. The nanopowder was recovered by freeze drying.

For preparation of scaffolds, the solvent (0.653 g), dispersant (0.265 g) and binder (0.040 g) were mixed together until the binder had fully dissolved. Next, with stirring

in an ultrasonic bath, each solid component was added sequentially in two or three aliquots, starting with the dried CGO nanoparticles (0.751 g), followed by the commercial CGO (1.047 g), and finally the polymer microbeads (0.153 g). The ultrasonication and stirring is effective in suspending the nanoparticles. This gave an ink with *ca*. 26 vol% solids content, which was an acceptable level for a screen printing ink.[11] In formulating the ink, two conditions were sought: high total solid content and high nanoparticle content. The order of addition was important to ensure that the nanoparticles were fully incorporated and homogenised, and the large differences in particle size between the different components helped to maintain an appropriate viscosity at high solid loadings.

The inks were screen printed onto 200 μm thick yttria-stabilised zirconia (YSZ) electrolytes. In the case of fuel cells, the procedure differed slightly between cells with different sintering temperatures for the anode, as shown schematically in Fig. 1. In the cells with scaffolds sintered at 1300 °C, one side was printed with the scaffold ink, while the other was printed with a thin layer of a CGO ink (this is to prevent reactions between the cathode material and the YSZ electrolyte). The printed pellets were then heat-treated at 1300 °C for one hour, with a 1 °C min^{-1} ramp rate during heating and cooling. The LSCF-CGO cathode (ink purchased from Fuel Cell Materials, Columbus, OH, USA) was then screenprinted, and the cell heat-treated at 1100 °C for three hours with a 5 °C min^{-1} ramp rate during heating and cooling. The process for the cells with

scaffolds heat-treated at 1000 °C was identical except the order of printing and sintering was a protective CGO layer first, followed by cathode and finally by the CGO scaffold. Cells with one, two or three layers of CGO scaffold were produced. The scaffold was dried in an oven at ~80 °C in between each printing cycle.

Next, the CGO scaffolds were impregnated with a nickel salt, again shown schematically in Fig. 1. The impregnation solution was prepared from 0.125 g of nickel nitrate hexahydrate in 200 μL of absolute ethanol. The solution was dropped onto the scaffold using a 1–20 μL pipette, with the volumes used determined empirically by visually observing whether the drop spread beyond the edge of the scaffold. If it did, then the volume of the following impregnation was reduced. The first volume used was determined by dropping pure ethanol onto the scaffold. In between each impregnation, the pellets were placed into a furnace directly at 500 °C for 5 min. This cycle was then repeated ten times. After the final impregnation, the pellets were placed into this furnace, and the temperature was increased to 800 °C at 5 °C min^{-1} and held for eight hours before cooling at 5 °C min^{-1}. The NiO loading was determined by weighing.

Four symmetrical cells were produced: a single layer cell heat-treated at 1300 °C; a single layer cell heat-treated at 1000 °C; a double layer cell heat-treated at 1000 °C and a triple layer cell heat-treated at 1000 °C. In addition, four button cells for fuel cell testing were produced from single

and double layer CGO scaffolds heat-treated at 1000 °C, and the same heat-treated at 1300 °C.

The scaffolds were studied by Ultra Small Angle X-ray Scattering (USAXS) to obtain information about the pore structure, while the *in situ* measurements allowed us to monitor the development of the microstructure with temperature.[25] For the USAXS measurements, single layered samples were printed onto MgO substrates (used instead of YSZ as it is less absorbing of X-rays) for analysis by USAXS, and heat-treated at either 1000 or 1300 °C. These samples were compared to a scaffold made from a "control" ink in which the nanoparticle component was replaced by extra commercial CGO. One sample printed on MgO was heat-treated *in situ*. The USAXS data were taken at sector 15, Advanced Photon Source, Argonne National Laboratory. Briefly, small angle scattering intensities around the direct beam was measured using channel cut Si crystal monochrometers in a Bonse–Hart configuration. The X-ray energy was 16.8 keV and the beam size was 1.5 × 0.4 mm. The sample was mounted in a LinkAm heater for the *in situ* USAXS measurements. The USAXS intensity is plotted as a function of the scattering vector $q = 4 \sin(\theta)/\lambda$ where λ is the X-ray wavelength and θ is the scattering angle. The USAXS spectra can be viewed as the autocorrelation of the electron density fluctuations inside the sample, resulting from contrast between the pores and the powders. The USAXS is normally plotted in a log–log plot, and at low q Guinier